

Yes. And this time I'm going to structure the **story of the problem itself** exactly the way you can later explain it in the 5-minute video.

I also want to be honest about the research: after this final pass, I would **not claim that “evidence graphs” or “correlated evidence” are themselves new**. Evidence graphs have existed in security/forensics for years, and recent 2026 work is using evidence graphs for other fraud/financial-fraud contexts.

What can be novel is the **specific payment-risk problem we define around evidence integrity and decision confidence**.

1. Where We Started — Problem Statement #1

“Payment fraud is evolving faster than traditional fraud detection systems.”

Our initial observation was straightforward.

Payment fraud systems generally learn from historical behavior:

- previous fraudulent transactions
- transaction amount
- device
- IP
- location
- merchant behavior
- velocity
- customer history
- behavioral patterns

The problem we identified was:

Fraudsters continuously change their behavior, while risk systems are inherently influenced by patterns observed in the past. This creates a gap between emerging fraud strategies and the behavior represented in existing risk models.

So our first problem statement was essentially:

How can we proactively discover new fraud strategies before they become real-world losses?

Why we rejected it

It sounded strong.

But our research immediately showed that this problem was **already heavily explored**.

Mastercard has a patent covering synthetic fraudulent transactions and a simulated authorization model specifically to evaluate vulnerabilities in payment authorization systems. The system generates synthetic transactions, enriches them with velocity information, runs them against a simulated authorization model and evaluates vulnerabilities.

There is also active research into adversarial fraud and fraud-model evasion.

And commercially, companies are already building AI-based fraud red-teaming systems.

Verdict

✗ REJECTED

We were trying to solve:

“How can we attack the fraud detector?”

But that problem already has substantial prior art.

2. Problem Statement #2

We then pushed the idea further.

Instead of merely discovering attacks, we asked:

What if the system could automatically find the best defensive change?

That gave us:

Adaptive Risk Policy Optimization

The problem became:

When a new fraud vulnerability is discovered, risk teams must decide how to modify their policies without creating excessive false positives or payment friction. How can AI discover the safest policy change that maximizes fraud prevention while minimizing legitimate-customer impact?

This was better.

Now we were solving:

Fraud discovery + defense optimization + customer experience.

Why we rejected it

Again, research killed it.

Commercial systems already address policy optimization, fraud-vs-friction trade-offs and simulated policy evaluation.

BetterBehave, for example, describes policy experimentation involving current versus simulated outcomes, fraud reduction, friction, false positives, policy variants and live experimentation.

And Darwinium's products already combine adversarial fraud testing with remediation/optimization concepts.

So:

“Find the best risk policy.”

was also not sufficiently differentiated.

Verdict

 **REJECTED**

3. Problem Statement #3

Then we went one layer deeper.

We noticed something important:

Risk systems don't just depend on the policy.

They depend on **signals**.

For example:

Device

IP

Location

Velocity

Merchant

Customer history

Payment method

Transaction amount

Behaviour

So we asked:

What happens when the signals themselves become unreliable?

That led to:

SignalTrust

The problem became:

Payment-risk signals do not remain equally reliable over time. Changes in user behavior, technology and attacker strategies can cause previously predictive signals to drift, weaken or become less useful. Risk systems therefore need to continuously determine which signals remain trustworthy and adjust their reliance accordingly.

This was significantly better.

Why we rejected it

Unfortunately, this also has substantial prior art.

A granted Mastercard-related patent, **US11410187B2**, explicitly describes detecting feature drift in fraud-prevention models and generating hardened models to mitigate that drift.

Recent fraud research is also heavily focused on dynamic graphs, concept drift, changing fraud behavior and time-sensitive fraud detection.

So:

“Detect when fraud signals become unreliable.”

is not enough.

Verdict

✗ REJECTED

4. Problem Statement #4

Then came the more interesting observation.

Suppose a payment has:

- trusted device
- trusted IP
- normal location

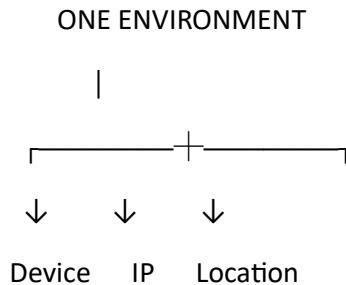
- normal merchant
- normal customer history

The risk engine might think:

“Five independent signals say this is safe.”

But are they really independent?

Consider:



The three signals might all originate from essentially the **same underlying circumstance**.

So the system may believe it has:

3 pieces of evidence

when it really has:

1 underlying source of evidence represented 3 times.

This led us to:

Evidence Independence

The problem became:

Risk systems combine multiple signals to increase confidence in a payment decision, but correlated or redundant signals can create false confidence. A system may appear to have strong multi-signal agreement even when most of its evidence originates from the same underlying entity, behavior or event.

This is much more interesting.

5. Then we attacked THIS idea too

And we found something important.

The concept of **evidence graphs themselves is not new.**

NIST research has used evidence graphs for representing dependencies between pieces of security evidence and reasoning about their relationships for more than a decade.

Recent work is also using evidence graphs in financial-fraud contexts. A 2026 paper on financial-statement fraud explicitly constructs evidence graphs with supporting and contradictory evidence.

Recent financial-transaction research also uses heterogeneous transaction graphs to capture relationships between accounts, devices, addresses and other entities.

Therefore:

“Let's make an evidence graph for fraud.”

would be absolutely insufficient.

Verdict

 **REFINED — NOT REJECTED**

Because we realized the graph is **not the invention**.

The real problem has to be higher-level.

6. The question that survived all of this

We finally arrived at the question:

“Can the risk decision itself be trusted?”

Not:

Is the transaction fraudulent?

Not:

Is this signal risky?

Not:

Has the model drifted?

Not:

Can an attacker bypass the model?

But:

How trustworthy is the evidence supporting the decision that the risk system just made?

That is a fundamentally different framing.

7. Final Problem Statement

The Problem

Modern payment-risk systems make real-time authorization decisions by combining numerous signals such as device, network, location, transaction behavior, merchant history and customer history. However, the presence of multiple signals does not necessarily mean that a decision is supported by multiple independent pieces of evidence. Signals can be correlated, redundant, contradictory, stale or influenced by the same underlying entity or behavioral event.

This can create a critical form of false confidence: a risk engine may assign high confidence to a decision because several signals agree, even though those signals provide little independent information. Conversely, genuinely independent evidence may be overlooked when the system treats all signals as equivalent.

The problem becomes even harder because the validity of evidence cannot always be known at decision time. Payment outcomes such as disputes, chargebacks or confirmed fraud may become available only later, meaning the system often cannot immediately determine whether the evidence supporting an earlier decision was actually reliable. Recent research specifically identifies delayed and incomplete fraud labels as a fundamental challenge in payment-fraud modeling.

Therefore, there is a gap between producing a risk decision and understanding the integrity of the evidence supporting that decision: existing risk systems primarily answer “What is the risk?”, while risk teams also need to understand “How trustworthy is the evidence behind this risk assessment?”

That is the problem I want us to own.

8. The Core Problem in One Sentence

If the evaluator remembers only **one sentence**, it should be:

“Payment systems can tell us how risky a transaction appears, but they don't necessarily tell us how trustworthy the evidence behind that risk assessment actually is.”

That's our problem.

9. Why this problem matters to Razorpay

Razorpay's Shield/Risk Engine already evaluates transactions using large numbers of signals and AI/ML-based risk analysis.

So we aren't saying:

“Razorpay doesn't have fraud detection.”

That would be a terrible pitch.

We're saying:

“You already have sophisticated risk intelligence. We identified another question that becomes increasingly important as risk systems become more complex: how much should we trust the evidence supporting each decision?”

That's a much more respectful and intelligent framing.

10. What exactly is the gap?

Think of the current world as:

SIGNALS



RISK MODEL



RISK SCORE



DECISION

The missing conceptual layer is:

SIGNALS



????????????????



RISK MODEL



RISK SCORE



DECISION

Our research identified that missing question as:

Evidence Integrity

11. What do we mean by “Evidence Integrity”?

We are **not** defining it as simply correlation.

That's important.

Evidence can fail in several different ways:

Dependency

Two signals may represent the same underlying source.

Redundancy

Several signals may provide almost no additional information.

Contradiction

One signal says safe while another strongly says suspicious.

Staleness

The evidence may have been reliable previously but no longer reflect current behavior.

Concentration

Most of the decision may depend on one evidence cluster.

Outcome uncertainty

We may not yet know whether the decision was actually correct.

Therefore:

Evidence integrity is the degree to which the evidence supporting a risk decision is independent, consistent, temporally relevant and ultimately accountable to observed outcomes.

That is the conceptual foundation.

12. The final problem we're actually solving

Putting everything together:

Final Problem Statement — Version for the project

Payment-risk systems increasingly rely on large numbers of real-time signals to make automated decisions, but a high number of agreeing signals can create an illusion of

certainty when those signals are correlated, redundant, contradictory, stale or derived from the same underlying source. At decision time, risk systems therefore may know the apparent risk of a payment without knowing the integrity of the evidence supporting that assessment.

This problem is compounded by delayed ground truth: whether a payment ultimately becomes fraudulent, disputed or legitimate may only become known after the original authorization decision. Consequently, risk teams lack a continuous way to evaluate not only whether their decisions were correct, but whether the evidence they relied upon was genuinely independent, reliable and predictive when the decision was made.

The resulting gap is an “evidence-confidence gap” between a risk score and the trustworthiness of the evidence supporting it. Closing this gap is important because increasingly sophisticated payment-risk systems can become confidently wrong when multiple correlated or compromised signals reinforce the same incorrect conclusion.

13. And this is the story for your 5-minute video

This gives you an **excellent research narrative** rather than pretending we magically invented an idea overnight:

Chapter 1 — We started with fraud evolution

“We initially wanted to proactively discover emerging fraud before conventional systems could detect it.”

Chapter 2 — We investigated prior art

“We found patents and commercial systems already addressing synthetic adversarial transactions, vulnerability testing and fraud-policy optimization.”

Chapter 3 — We went deeper

“We then looked at signal reliability and discovered that feature drift and adaptive fraud-signal weighting were also established areas with patents and research.”

Chapter 4 — We found the deeper issue

“We realized the fundamental question wasn't simply whether a signal was risky. It was whether multiple signals actually represented independent evidence.”

Chapter 5 — We attacked that idea too

“Evidence graphs and correlated evidence analysis already exist in security and fraud research, so we deliberately didn't stop there.”

Chapter 6 — The final gap

“We found a gap between a risk score and the trustworthiness of the evidence supporting that score — especially when evidence is correlated, contradictory, stale, concentrated, or later invalidated by the actual payment outcome.”

Final line

“So we stopped asking: ‘Is this payment risky?’ and started asking: ‘Can we trust why the system thinks it is risky?’”

That is the problem statement I would lock.

And importantly, our research journey itself becomes evidence of maturity: we didn't settle for the first cool idea—we actively tried to disprove our own idea until we found a problem worth solving.